COURSE DESCRIPTION FORM

INSTITUTION FAST NUCES

PROGRAM (S) TO BE EVALUATED Computer Science

A. Course Description

Course Code	CS 3001
Course Title	Computer Networks
Credit Hours	**3 + 1**
Prerequisites by Course(s) and Topics	CS2001-Data Structures
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	Mid-1: 15 Mid-2: 15 Quiz: 5 (minimum 2 quizzes, 2.5 each) Assignment: 5 (minimum 2 Assignments, 2.5 each) Project: 10 Final: 50
Course Coordinator	
Class Teacher	
URL (if any)	
Current Catalog Description	The learning and skill-based objectives of this course resolve around the following questions: ● How does the global network infrastructure work and what are the design principles on which it is based? ● In what ways are these design principles compromised in practice? ● How should Internet applications be written, so they can obtain the best possible performance both for themselves and for others using the infrastructure? ● How do we ensure that it will work well in the future in the face of rapidly growing scale and heterogeneity? The course will focus on the design & undergraduate level analysis of large-scale networked systems and tool (Wireshark, packet tracer) based implementation and evaluation of small-scale networked systems in the Lab.
Textbook (or **Laboratory Manual** for Laboratory Courses)	J. F. Kurose and K. W. Ross --- **Computer Networking: A Top-Down Approach, 9 Edition** Wireshark, Packet Tracer, and GNS3 Labs (mostly by the lab instructors)
Reference Material	A. S. Tannenbaum and D. J. Wetherall --- **Computer Networks, 6th Editicon**

	B. A. Forouzan --- **Data Communications & Networking with TCP/IP Protocol Su** **6th Edition** W. Stallings --- **Data and Computer Communications, 10th Edition**

A. Course Learning Outcomes (CL

No.	Course Learning Outcomes (CLO)	Domain	Taxonomy Level	PLO	Too
01	Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack. (2) (2)	Cognitiv e	C2 (Describe)	2	Q, M, F
02	Demonstrate the basics of network concepts using state-of-the-art network tools.(3) (2)	Cognitiv e	C3 (Apply)	2	Q, M, F
03	Demonstrate operations of various classical routing and switching protocol via simulators.(3) (4)	Cognitiv e	C3 (Apply)	4	Q, M, P,

Tool: A = Assignment, Q = Quiz, P = Project, M = Mid-term, F=Final (End-te

B. Graduating Attributes -GAs (Program Learning Outcomes – PL

1	Academic Education	Completion of an accredited program of study designed to prepare graduates as computing professionals
2	Knowledge for Solving Computing Problems	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Problem Analysis	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
4	Design/ Development of Solutions	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
5	Modern Tool Usage	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations
6	Individual and Team Work	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings
7	Communication	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions
8	Computing Professionalism and Society	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice
9	Ethics	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
10	Life-long Learning	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

	C. Relation between CLOs and G (CLO: Course Learning Outcome, PLOs: Program Learning

		Graduating Attributes- GAs or PLOs									
		1	2	3	4	5	6	7	8	9	10
CLOs	1		✔								
	2		✔								
	3				✔						

Topics Covered in the Course, with Number of Lectures on Each Topic (assume 15-week instruction and one-hour lectures)

Week	Duration	Topics Covered	C
1.	L1 = 1 hour L2 = 1 hour L3 = 1 hour	1.1 - Introduction, Course 1.2 - Network Edge, 1.3 - Network Core (Tiers of ISPs, Backbones)	
2.	L1 = 1 hour L2 = 1 hour L3 = 1 hour	1.4 - Delay, Loss and Throughput 1.5 - Protocol Layers, Service Model, 1.6 - Network Under Attacks 1.7 History of Computer Networking and the Internet 2.1 - Principles of network Applications	
3.	L1-2 = 2 hours L3 = 1 hour	2.2 - Web and HTTP 2.3 - Electronic Mail`	
4.	L1-2 = 2 hours L3 = 1 hour	2.4 - DNS—The Internet's Directory Service 2.5 - Video Streaming and Content Distribution Networks	1
5.	L1 = 1 hour L2 = 1 hour L3 = 1 hour	3.1 - Transport Layer service 3.2 - Multiplexing and De-multiplexing 3.3 - Connectionless Transport UDP	1
6.	**1 Hour** L1 = 1 hour	**Midterm # 1** 3.4 – Principles of Reliable data transport	
7.	L1 = 1 hour L2 = 1 hour L3 = 1 hour	3.5 - Connection Oriented Transport: TCP 3.5.2 TCP Segment Structure 3.5.3 Round-Trip Time Estimation and Timeout 3.5.4 Reliable Data Transfer 3.5.5 Flow Control 3.5.6 TCP Connection Management ***(Project will be announced on the 7th week)***	
8.	L1-L3 = 3 hours	3.7 - TCP Congestion Control	

- No

9.	L1 = 1 hour L2 = 1 hour L3 = 1 hour	4.1 - Network Layer- Data Plane Introduction 4.2.1 - Destination-Based Forwarding 4.3.1 - Internet Protocol (IPv4 Datagram Format) 4.3.2 - IPv4 Addressing	
10.	L1 = 1 hour L2 = 1 hour L3: 1 hour	4.3.2 - IPv4 Addressing Class C Subnetting Variable Length Subnet Mask (VLSM) 4.3.3 - Network Address Translation (NAT) 4.3.4 - IPv6 4.4 - Generalized Forwarding and SDN	1
11.	**1 Hour** L1 = 1 hour	**Midterm # 2** 5.1 - Network Layer-Control Plane Overview 5.2 - Routing Algorithms (Link State)	
12.	L1-L2 = 2 hours L3 = 1 hour	5.2 - Routing Algorithms (Distance Vector) 5.3 - Intra-AS routing in the Internet: OSPF 5.4 - Routing Among the ISPs: BGP	1
13.	L1-L3 = 3 hours	5.4 - Routing Among the ISPs: BGP 5.5 The SDN Control Plane	1
14.	L1 = 1 hour L2-L3 = 2 hours	6.1 - Introduction to Link layer 6.2 - Error-Detection & Correction Techniques 6.3 – Multiple Access Links and Protocols	
15.	L1-L2 = 2 hours L3 = 1 hour	6.4 Switched Local Area Networks 6.6 - Data Center Networking	
16.	3 hours	Course wrap-up	1

Laboratory Projects/Experiments Done in the Course	Project is focused on the application of network fundamentals and practices to deve efficient networking solutions and applications.			
Programming Assignments Done in the Course	Socket Programming			
Class Time Spent on (in credit hours)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethic Issues**
	40%	30%	30%	0%
Oral and Written Communications				

ReQuizz and Late Assignment submission

- Usage of generative artificial intelligence (AI) such as ChatGPT or Microsoft Copilot for producing assessment submissions etc. should be mentioned clearly with reference.

Instructor Name: ______________

Instructor Signature: ________________

Date: